An Application of Established Computational Models of Estimation in Three Real-World Domains

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Abstract

Research on quantitative estimation often relies on either formal computational models that are tested with artificial stimuli or informal theories applied to real-world tasks. The current study bridges these approaches by testing established computational models of estimation with naturalistic stimuli from three domains: food, countries, and mammals. Participants were trained to estimate domain-specific criteria (grams of carbohydrates per 100g for food items, life expectancy for countries, and days until female maturity for mammals) and subsequently estimated the criterion values for new objects in a testing phase. We compared five candidate models - the Cue Abstraction Model, the Generalized Context Model, the RulEx-J hybrid, the Mapping Model, and QuickEst Heuristic - using a simulation-based Bayesian model comparison approach. Results showed that all models except QuickEst accounted for participants' estimates, but dominant processes varied by domain. These findings highlight domain-specific variability in estimation strategies while demonstrating that computational models originally developed with artificial tasks can successfully capture judgments in real-world domains. Our results underscore the importance of testing cognitive process models beyond controlled laboratory contexts to better understand the mechanisms underlying everyday quantitative estimation.

Keywords: quantitative estimation, quantitative judgment, multidimensional scaling, cognitive modeling, Bayesian model comparison

Introduction

In a classic joke, a physicist proposes a method for increasing the milk production of a dairy farm that only works "in the case of spherical cows in a vacuum". This joke illustrates a recurring dilemma in science: to explain complex phenomena, researchers often must rely on abstractions and simplifying assumptions. This trade-off between simplifying assumptions necessary for theory testing and realistic conditions necessary for theory application can also be found in research on quantitative judgments and estimation. Here, either computational models, which are precise implementations of the underlying cognitive processes, are tested in simplified laboratory environments with artificial stimuli (e.g., Bröder & Gräf, 2018; Hoffmann et al., 2016; Juslin et al., 2003; Pachur & Olsson, 2012; von Helversen & Rieskamp, 2009a) or somewhat imprecise theories are used together with realistic stimuli (e.g., Bröder et al., 2023; Brown & Siegler, 1993; Wohldmann, 2015).

The present work aims to bridge this gap. We compare established computational models of estimation using naturalistic stimuli in three domains: (1) a biological taxonomic domain (mammals), (2) a geographic/social domain (countries), and (3) a highly relevant health-related domain (food). This approach allows us to test whether the computational models of the assumed underlying cognitive processes of people's estimates, which were mainly developed and tested with artificial stimuli, are able to account for participants' estimates in these real-world domains. More specifically, we ask which processes best describe participants' estimates in each domain, whether similar processes operate in all domains, or whether the processes used are content-dependent.

Following this introduction, we first provide a brief overview of research on quantitative estimation in general. We then describe how the computational models developed in this line of research can be applied to estimates of naturalistic stimuli, setting the stage for our modeling approach used in this paper. Finally, we introduce the set of candidate models of estimation that we compare in this study in detail.

Background: Research on Quantitative Estimation

In an *estimation* task (or, more broadly, a quantitative judgment task), people assign a symbolic representation of a quantity (typically a number) to an object or set of objects to express their belief about a certain attribute (the judgment or estimation *criterion*). Such estimates are a routine part of daily life: We may estimate the time to get to the office, the number of people following our party invitation, or the calories in a dessert.

Research on the cognitive processes underlying such estimates has basically followed three different traditions. At the one end of the spectrum of degree of formalization is Brown and Siegler's (1993) highly influential work dealing with real-world quantitative estimation, involving estimates of country populations (Brown, 2002; Brown & Siegler, 1993), other geographical attributes (Brown & Siegler, 2001; Friedman & Brown, 2000), car prices (Murray & Brown, 2009), college tuition rates (Lawson & Bhagat, 2002), nutritional values (Groß et al., 2024; Wohldmann, 2015), and carbon footprints of food items (Bröder et al., 2023). Brown (2002) summarized much of this research in the theoretical account known as the *metrics* and *mapping* account of estimation. According to this framework, accurate estimation relies on two kinds of domain-specific knowledge: Metric knowledge, which captures distributional properties of the criterion (e.g., minimum, maximum, median, or range), and mapping knowledge, which reflects the rank order of objects with respect to the criterion. This framework has been very fruitful in explaining estimation and especially its improvement through a simple procedure called *seeding*, whereby providing criterion information about a small sample of objects can massively improve the estimation accuracy for new objects from the same domain (e.g., Brown & Siegler, 1993). However, while the framework has been tested in real-world domains, it has not yet been cast in precise computational terms (but see von Helversen and Rieskamp, 2008, discussed below).

The middle ground is covered by the Brunswikian tradition (Brehmer, 1994; Hammond, 1955; Karelaia & Hogarth, 2008), in which judgments of real-world stimuli are modeled by linear regression models using cues as predictors. Although this adds valuable additional knowledge to the Brown-Siegler approach, the approach has hitherto mainly been limited to these linear models and close relatives thereof (Einhorn et al., 1979).

At the other end of the spectrum, and the largest research tradition in the last decades, is the use of computational models that specify the cognitive processes underlying peoples estimates as functions operating on object *attributes*, also called *features* or *cues* (e.g., Bröder et al., 2010; Hoffmann et al., 2013, 2014, 2016; Izydorczyk & Bröder, 2021, 2022; Juslin et al., 2003; Karlsson et al., 2008; Pachur & Olsson, 2012; Rosner & von Helversen, 2019; Seitz et al., 2025). For instance, linear integration models inspired by the Brunswikian tradition attach a weight to each attribute of an object and compute a weighted sum as the judgment of the criterion (Einhorn et al., 1979, Juslin et al., 2003; see Brehmer, 1994 for an overview). According to these models, the calorie content of a dessert might be computed if the amounts of carbohydrates, fat, and proteins (as the cues) were known and assigned the appropriate weights. Other models that have been cast in computational terms (detailed below) also rely on cues: *Exemplar models* (e.g., Juslin et al., 2008; Nosofsky, 1984) assume a similarity match of a to-be-judged probe with exemplars in memory. Here, similarity is defined as a function of the features that form the dimensions of a psychological space in which stimulus dissimilarity is represented as a spatial distance. Finally, the *Mapping Model* by von Helversen and Rieskamp (2008, 2009a, 2009b, inspired by Brown and Siegler's framework) and the *QuickEst* heuristic (Hertwig et al., 1999; Hertwig & Hoffrage, 2001) assume that stimuli are grouped or ordered by their feature patterns to obtain a final estimate. Since testing these computational models relies on the experimenter's insight into the functional stimulus features or cues, most studies have used artificial or, at most, semi-realistic stimuli for model tests. Examples are the well-known "death bugs" (Juslin et al., 2008; Trippas & Pachur, 2019; von Helversen & Rieskamp, 2008), artificial butterflies (Hoffmann et al., 2016), artificial flowers (Izydorczyk & Bröder, 2022), fictitious job candidate profiles (Scholz et al., 2015; von Helversen, Karlsson, et al., 2014), fictitious patients' symptom patterns (Bröder et al., 2010; Persson & Rieskamp, 2009), geometric shapes (Donkin et al., 2015; Goldstone, 1994; Seitz et al., 2025), or cartoon characters (Bröder et al., 2025; Hoffmann et al., 2014), to name but a few.

As a pointed summary of cognitive research on quantitative judgment, one may thus conclude that (1) either relevant real-world stimuli have been explained with somewhat imprecise theories, or (2) precise computational models have been used to explain estimation with arguably irrelevant artificial stimuli. Importantly, as experimental psychologists, we do not devalue controlled artificial stimuli as a valid means to test cognitive process models, as it is often necessary to eliminate nuisance variables for precise theory testing. Carefully designed laboratory experiments are indispensable for isolating mechanisms and deriving clear theoretical predictions. However, for cumulative theoretical development in psychology, it is equally important that theories prove robust beyond the tightly controlled conditions under which they were developed. As a science ultimately concerned with explaining and predicting behavior in the real world, psychology must therefore examine whether cognitive process models generalize to ecologically more valid settings (Brunswik, 1955). As a second step, we consider it worthwhile to investigate whether theories that stood the test of controlled laboratory situations also generalize to real-world stimuli (see Goldstein and Hogarth, 1997). The goal of the current research is, therefore, to compare computational models in the realm of naturalistic stimuli.

Extracting Features from Naturalistic Stimuli

Given that all computational models of judgment rely on stimulus features as input to cognitive processes, a crucial step in applying them to real-world domains is to determine how naturalistic stimuli can be represented in terms of psychologically meaningful dimensions. Unlike in artificial laboratory tasks, the relevant features of real-world objects are not predefined, and it is often unclear which of them are functional in cognitive tasks. Researchers may define them ad hoc, according to their subjective beliefs of relevance, or by asking for participants' introspection. Since all these approaches run the risk of missing important cues, in this work, we relied on another method originally developed by Nosofsky and colleagues (2018, 2018, 2020, 2025) in the natural domain of minerals. In this approach, participants provide pairwise similarity ratings for stimuli within a domain. These are then subjected to a *multidimensional scaling* (MDS) analysis, which leads to a spatial representation of the stimuli in a high-dimensional space where similar stimuli are located closer together than dissimilar stimuli (Hout et al., 2013; Shepard, 1962). The optimal number of dimensions m of this spatial representation can be determined by various methods, and all stimuli are then represented by an m-dimensional vector of coordinates. These can be interpreted as feature values and form the input to computational models of estimation. In a toy environment using a set of $n = 32$ bird species, Izydorczyk and Bröder (2024) validated the method for estimation models by showing that a computational model using the MDS-extracted bird features replicated a well-known effect that had been demonstrated with artificial stimuli before (Trippas & Pachur, 2019).

In a companion paper to the current article (Izydorczyk & Bröder, 2025), we applied this method to sets of 80 representative items of three domains (a) countries, (b) mammals, and (c) food items. For all 3,160 paired comparisons in each domain, we obtained similarity ratings from more than 500 participants per domain on a 7-point scale (1 = not similar, 7 = very similar), which were then subjected to a non-metric MDS analysis. Based on leave-one-out cross-validation, we determined the optimal number of dimensions to be 10 for countries, 10 for mammals, and 14 for foods. We refer the reader to Izydorczyk and Bröder (2025) for further procedural details.

In addition, as a proof-of-relevance, Izydorczyk and Bröder (2025) had participants judge a quantitative criterion for all stimuli in each domain: life expectancy for countries, time to maturity in days for female mammals, and calories for food items. In each domain, the extracted features linearly explained more than 65% of the criterion variance, demonstrating that the extracted features contain useful information about the stimuli. Also, linear regressions of participants' judgments showed that the features explained 37%, 57%, and 44% of variance on average in the country, mammals, and food domains, respectively, demonstrating that the features are functional in explaining the estimates as well. Hence, we are confident that there is a solid basis for comparing different computational models in their ability to explain these estimates.

In the next section, we specify the candidate models before turning to the design of our judgment study and the modeling results.

Models of Judgment and Estimation Processes

In this section, we introduce the candidate models of quantitative estimation compared in this study. These models come from different research traditions and differ quite substantially in the type of information learned and abstracted from previously encountered objects and in how they process a new object to generate an estimate for this object. Specifically, we introduce exemplar-based and rule-based models to quantitative judgment, the RulEx-J model, the Mapping model, and the QuickEst heuristic.

Across all models, we adopt a common notation: Each model predicts the estimated criterion value $\hat{y}_p$ for a probe p (the novel object to be judged) based on information learned from a set of k previously encountered objects (the exemplars). Both the probe and each exemplar are described by a set of m cues, attributes, or features.

The Rule model

Based on Brunswik's lens model (Brunswik, 1955), rule-based process models assume that people combine and integrate cue information according to a learned rule (Hoffmann et al., 2019). Most commonly, this rule is assumed to be a weighted linear additive function (Brehmer, 1994; Juslin et al., 2003, 2008), because people are quite good at learning such rules, show a preference for using them, and they have been shown to fit judgment data across a wide range of domains (Ashby et al., 2001; Brehmer, 1974, 1994; Einhorn et al., 1979; Fischbein et al., 1985; Hoffmann et al., 2014, 2018; Kalish et al., 2004; Karelaia & Hogarth, 2008; Lagnado et al., 2006; McDaniel & Busemeyer, 2005).

Juslin et al. (2003) formalized the assumption of an additive linear integration of multiple cues in the ***C****ue* ***A****bstraction* ***M****odel* (CAM), which predicts an object's p estimated criterion value $\hat{y}_p$ as a weighted sum of m cue values:

$$\hat{y}_p = \beta_0 + \sum_{i=1}^{m} \beta_i \times \text{cue}_i, \tag{1}$$

where β_0 is the intercept and β_i are the learned cue weights associated with the m cues. This rule-based model is quite flexible and does not necessarily imply a compensatory processing of all cues, but can also mimic simpler heuristics (e.g., lexicographic rule) when some weights are near zero (Bröder, 2000; Gigerenzer & Goldstein, 1996, 1999).

The Exemplar model

Exemplar models originate from research about categorization (Medin & Schaffer, 1978; Nosofsky, 1984), but are also successfully used to account for continuous judgments (e.g., Juslin & Persson, 2002). Exemplar models assume that people store previously encountered objects (the exemplars) along with their criterion values or category memberships as separate traces in episodic long-term memory (Estes, 1986; Hintzman, 1986; Medin & Schaffer, 1978). When a new, to-be-judged object is encountered, it serves as a retrieval cue that activates similar exemplars from memory. The judgment is then formed by integrating the criterion values of these retrieved exemplars, where more similar exemplars exert greater influence on the final judgment. Hence, the final judgment is a similarity-weighted average of the criterion values associated with the retrieved exemplars. Exemplar models have been shown to account quite well for participants' quantitative judgments, especially when cue–criterion relationships are complex and non-linear (Bröder et al., 2010; Hoffmann et al., 2013; Izydorczyk & Bröder, 2021; Juslin & Persson, 2002;

Juslin et al., 2003; Mata et al., 2012; Pachur & Olsson, 2012; Rosner & von Helversen, 2019; von Helversen & Rieskamp, 2008; Wirebring et al., 2018).

One formalization of an exemplar-based model is the ***G**eneralized **C**ontext **M**odel* (GCM, Nosofsky, 1986, 2011) extended to account for quantitative estimates (Albrecht et al., 2019; Hoffmann et al., 2016; Izydorczyk & Bröder, 2022; Seitz et al., 2025). In the GCM, each exemplar is represented as a single point in an m-dimensional psychological space. The distance d_{pe} between the to-be-judged object (i.e., the probe) p and an exemplar e, both represented by a set of m cues, is then given by the weighted Minkowski distance metric,

$$d_{pe} = \sqrt[r]{\sum_{i=1}^{m} w_i \cdot |p_i - e_i|^r}, \tag{2}$$

where r determines the form of the distance metric, with $r = 1$ (city-block distance metric) typically used for highly separable-dimension stimuli and $r = 2$ (Euclidean distance metric) for integral-dimension stimuli (Garner, 1974; Grau & Nelson, 1988; Lee, 2008; Shepard, 1991). Throughout this article, we will assume a fixed $r = 2$. The w_i values are freely estimated attention-weight parameters (with $0 \leq w_i \leq m$, and $\sum w_i = m$), reflecting the degree of attention that observers give to each dimension or cue i. Based on Shepard's law of generalization (Shepard, 1957, 1987), the similarity between the probe and exemplar is given by

$$s_{pe} = \exp(-h \cdot d_{pe}), \tag{3}$$

where h is the sensitivity parameter that governs how rapidly similarity declines with increasing distance. When h is large (i.e., steeper similarity gradient), exemplars become less similar to distant stimuli more quickly, and thus responses are based primarily on exemplars nearest to the probe. When h is small, more distant exemplars are integrated into the final response. Finally, the predicted estimated criterion value of the probe $\hat{y}_p$ is computed as the similarity-weighted average of the criterion values of all k exemplars:

$$\hat{y}_p = \frac{\sum_{i=1}^{k} s_i \cdot crit_i}{\sum_{i=1}^{k} s_i}. \tag{4}$$

Blending model RulEx-J

Traditionally, rule- and exemplar-based models have been treated as distinct processes, assuming individuals rely on only one of them at a time (Juslin et al., 2003, 2008; Karlsson et al., 2008; Pachur & Olsson, 2012; von Helversen et al., 2010). However, empirical findings from judgment and categorization research suggest that both processes may operate in parallel and interact (Allen & Brooks, 1991; Brooks & Hannah, 2006; Erickson & Kruschke, 1998; Hahn et al., 2010; Hannah & Brooks, 2009; Macrae et al., 1998; Regehr & Brooks, 1993; Rosner & von Helversen, 2019; Thibaut et al., 2018; von Helversen, Herzog, & Rieskamp, 2014). People appear to blend rules and exemplars when making judgments, with the relative influence of each process depending on task demands, stimuli, or environmental structure (e.g., Albrecht et al., 2019; Bröder et al., 2017; Wirebring et al., 2018). Based on this evidence, the *RulEx-J* model (Bröder & Gräf, 2018; Bröder et al., 2017; Izydorczyk & Bröder, 2022) proposes a continuous integration of rule-based and exemplar-based processes, with a stimulus being processed by both a rule module and an exemplar module in parallel. The final judgment $\hat{y}_p$ for the probe is a weighted combination of the two interim judgments J_R and J_E from the respective rule- or exemplar-based processes described in Equations 1-4, weighted by the parameter α:

$$\hat{y}_p = \alpha \times J_R + (1 - \alpha) \times J_E, \tag{5}$$

The α parameter measures the relative contribution of the rule- and exemplar-based process on the final judgment. It can range from 0 to 1, with larger values indicating more rule-based processing. Thus, for $\alpha = 0$ or $\alpha = 1$, RulEx-J contains the exemplar and the rule model as special cases, respectively. While several models in the categorization and judgment literature have proposed alternative blending or mixture mechanisms (e.g., Albrecht et al., 2019; Anderson & Betz, 2001; Love et al., 2004; Nosofsky et al., 1994), the RulEx-J model is currently the most widely used hybrid model in continuous judgment research (Bröder & Gräf, 2018; Izydorczyk & Bröder, 2022; Seitz et al., 2025) and, unlike the CX-COM model (Albrecht et al., 2019), does not depend on a specific stimulus and exemplar selection, making it broadly applicable across different tasks[1].

Mapping Model

Based on Brown and Siegler's (1993) work on the estimation of real-world quantities and inspired by Gigerenzer et al.'s (1999) simple heuristics framework, von Helversen and Rieskamp (2008, 2009a) introduced the *Mapping model* as a cognitive theory for quantitative estimation. The Mapping model uses 0/1-binary cues that are positively correlated with the criterion value; that is, objects with a positive cue value (i.e., cue = 1) tend to have higher criterion values on average than those with a negative cue value (i.e., cue = 0). According to the Mapping model, people estimate an object's criterion value in two steps. First, they categorize the object via a simple counting rule, where the category is determined by the sum of all positive cue values across m binary cues:

$$c_p = \sum_{i=1}^{m} \mathrm{cue}_i \tag{6}$$

Here, c_p is the cue sum category of the probe p. Each cue sum category is then associated with a typical criterion value, computed as the median of all previously encountered objects

[1] The predictions of CX-COM (combining Cue abstraction with eXemplar memory assuming COMpetitive memory retrieval) and RulEx-J can be quite indistinguishable for standard sets of stimuli and experimental design that are not explicitly selected to test the models against each other, we thus decided to not include the CX-COM model

that are in the same category (i.e., have the same cue sum). The estimated criterion value for a probe is then this typical criterion value for the corresponding category, that is, the median criterion value of all exemplars e with the same cue sum:

$$\hat{y}_p = Median(crit_e,\, c_e = c_p) \tag{7}$$

Thus, all objects with the same cue sum are predicted to have the same estimated criterion value.

Since our study uses continuous rather than binary cues, we implemented an adapted version of the Mapping model for continuous cues, following the approach of von Helversen and Rieskamp (2009b). Similar to the original model, the first step is to determine the latent category structure. Instead of computing cue sums across binary cues, we calculated the *average* cue value for each learned exemplar e across all m continuous cues - after recoding them, if necessary, to ensure all cues were positively correlated with the criterion value:

$$\bar{c}_e = \frac{1}{m}\sum_{i=1}^{m} \mathrm{cue}_{ei} \tag{8}$$

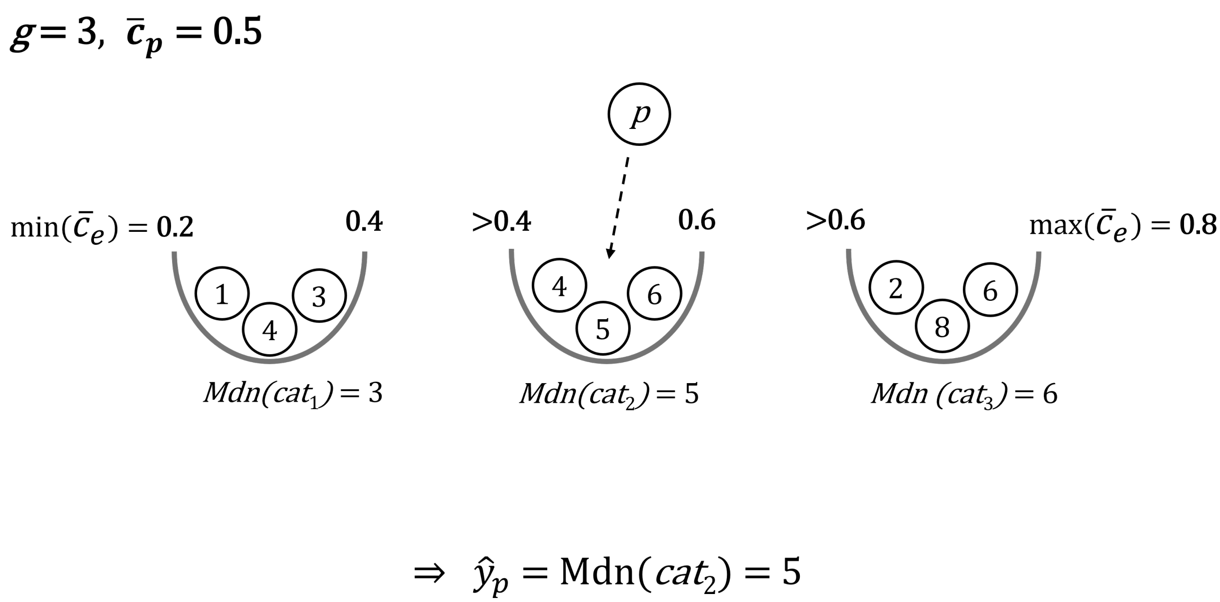
Based on these averages, we created g equally spaced category bins on the range defined by the minimum and maximum average cue value of all exemplars (i.e., $min(\bar{c}_e)$ and $max(\bar{c}_e)$), where g is a free parameter that could range from 2 to the number of exemplars m[2]. This is illustrated in Figure 1, for $g = 3$, and exemplary minimum and maximum average cue values of $min(\bar{c}_e) = 0.2$ and $max(\bar{c}_e) = 0.8$. As in the original Mapping model, each category is then associated with a typical criterion value, computed as the median of the criterion values of all exemplars in this category. The final predicted criterion value for a probe is then

[2] von Helversen and Rieskamp (2009b) fixed this parameter to 7. In our approach, we will estimate it as a free parameter.

determined by the typical criterion value of the corresponding category. For instance, the probe with the average cue value of $\bar{c}_p = 0.5$ in Figure 1 falls into the second category bin and thus has a predicted criterion value of $\hat{y}_p = 5$.

Figure 1

Illustration of the Mapping model for continuous cues with $g = 3$ categories.



Note. Balls represent the exemplars within each category and the numbers their corresponding criterion values. g = number of latent categories. $\bar{c}_p$ = average cue value of the probe p. $\hat{y}_p$ = predicted criterion value of the probe p.

If no exemplars were present in that category, we used the average of the adjacent categories' typical values. If the probe's average cue value fell outside the observed exemplar range, we assigned it to the closest category at the boundary.

QuickEst

Hertwig and colleagues (2001, 1999) proposed a fast and frugal heuristic for quantitative estimation, called *QuickEst*, which relies on a small number of cues. As in the original formulation of the Mapping model, these cues are assumed to be binary and are coded to correlate positively with the criterion. According to the QuickEst-heuristic, people have knowledge about the average criterion value for objects with positive cue values (conditional positive mean) and those with negative cue values (conditional negative mean). For example, they might know the average population size of cities that are state capitals versus those that are not. Cues are processed sequentially in an order determined by their conditional negative means, starting with the cue that has the smallest such mean. Search terminates as soon as a cue is encountered for which the object lacks the corresponding property (i.e., the first negative cue). The estimate is then the conditional negative mean of that cue, rounded to the nearest spontaneous number—a number in the form $a \cdot 10^x$, where $a \in 1, 1.5, 2, 3, 5, 7$. For example, 300, 500, 700, and 1000 are spontaneous numbers, but 900 is not.

Since the original QuickEst heuristic relies on binary cues, and the cues we use are continuous, we first recoded our cues to ensure they were all positively correlated with the criterion value. We then discretized them by assigning a value of 1 to all cue values above the mean and 0 to those below, to stay with the original implementation of QuickEst. As in the original formulation, cues are ordered based on their conditional negative mean — that is, the average criterion value of all exemplars with a value of 0 on that cue. These average values are then rounded down to the nearest multiple of ten (e.g., 856 -> 850). On each trial, the binary cues of the current item are checked in that order. The final estimate is determined by the conditional negative mean (rounded to tens) of the first cue for which the current object has a value of 0. If the object has a value of 1 for all cues, the estimate defaults to the maximum criterion value observed in the exemplars (i.e., a "catch-all" prediction). This implementation does not use spontaneous numbers, as in the original QuickEst formulation, but rounds to the nearest tens place instead. Moreover, we assume uncertainty in memory for the criterion values of the exemplars, where the remembered criterion value for each exemplar is drawn from a normal distribution centered on its true value, with some person-specific precision σ_{QEst}.

To our knowledge, direct empirical tests of QuickEst as a cognitive model have been rare. In two experiments, Hausmann et al. (2007) did *not* observe response time distributions or information search as predicted by QuickEst. von Helversen and Rieskamp (2008) compared QuickEst to the Mapping model, the Rule model, and an Exemplar model, finding worse model fits for QuickEst in most cases. Both studies, therefore, question the validity of the heuristic as a cognitive model. However, the latter study and Experiment 2 of Hausmann et al. (2007) used artificially constructed cue profiles (death bugs, fictitious patients, fictitious cities), and therefore, we nevertheless included QuickEst here as a competitor since it has been explicitly formulated as a model inspired by the *ecological rationality* idea proposed by Gigerenzer and Goldstein (1999). This approach assumes that our cognitive strategies are tailored to natural environments, which may be misrepresented in experiments with artificial stimuli.

The Present Study

In this study, participants were trained to estimate criterion variables from three real-world domains: food, countries, and mammals. Using multidimensional scaling (MDS)-based features to represent the 80 stimuli in each domain, we compared five computational models of quantitative judgment in their ability to account for participants' estimates. We examined which process model best describes participants' estimates, how well the models capture individual estimation behavior, whether similar processes operate across domains or are content-dependent, and how well the models generalize to novel stimuli. Details on the stimuli and extraction of MDS-based features are provided in Izydorczyk and Bröder (2025)[3]; a brief summary is provided below. This design tests whether computational models developed for artificial laboratory tasks can also capture human estimation in naturalistic domains.

[3] The companion paper focuses on the creation of the stimulus materials, pairwise similarity ratings, and MDS analysis, and includes parts of the current data as a proof-of-relevance demonstration, but no cognitive modeling of estimation data. Thus, all analyses presented here are original.

Method

Design and Procedure

We used a typical multiple-cue judgment experiment procedure consisting of a training phase followed by a testing phase (e.g., Hoffmann et al., 2016; Juslin et al., 2003). After providing informed consent, participants were trained in each domain to estimate a specific criterion: the grams of carbohydrates per 100g for food items, life expectancy for countries, or the number of days until female maturity for mammals. During the training phase, participants completed up to 10 blocks of training in which they estimated the criterion values of $k = 12$ exemplars and received immediate feedback about the true values after each response. Exemplars were selected through a simulation procedure aimed at identifying a stimulus set that maximizes discriminability between rule-based and exemplar-based models (the two most widely used and cited estimation processes in the literature) while also covering a broad range of criterion values (a list of all items and exemplars can be found in Tables A1-A3). Training lasted a minimum of five blocks and continued until participants met an accuracy criterion: at least 10 out of 12 estimates falling within $\pm$ 10% of the true value. Upon reaching this threshold or after completing 10 blocks, participants proceeded to the testing phase, where they estimated the criterion for all 80 stimuli (12 old exemplars and 68 new stimuli) in randomized order without feedback.

In addition, three attention-check trials were included in the testing phase. In these trials, a basketball was presented instead of a domain-specific stimulus, and participants were instructed to respond with "999". To get an idea about people's prior knowledge of the domain, at the end of the experiment, participants rated their general knowledge of the respective domain on a 7-point scale (1 = very low, 7 = very high). They also answered an open-ended question asking them to describe the strategy they used when making their judgments. In addition, participants provided demographic information and indicated whether their data should be included in the analysis or excluded Aust et al. (2013). Finally, participants reported whether they had used any form of aid (e.g., note-taking, internet searches), experienced any technical difficulties, and were able to complete the study without interruption.

The experiment, the full materials, and the preregistration are available at the Open Science Framework (OSF, https://osf.io/4m25d/overview?view_only=22adcafffca34c1f8bd10d9835cea98f; Izydorczyk and Bröder, 2026)

Materials

As stimuli for each domain, we used the 80 items presented in Izydorczyk and Bröder (2025), which were selected to capture broad and representative variation within each domain, for instance, foods from different categories (e.g., fruits, vegetables, meat, and sweets), countries from different continental areas, and mammals from a wide range of taxonomic categories. Figure 2 shows two example stimuli from each domain. A complete overview of all stimuli and their corresponding criterion values can be found in Tables A1-A3 in Appendix A. All stimuli and corresponding images used in the experiment are available online (https://osf.io/4m25d/overview?view_only=22adcafffca34c1f8bd10d9835cea98f) under a CC-BY-NC-SA-4.0 license.

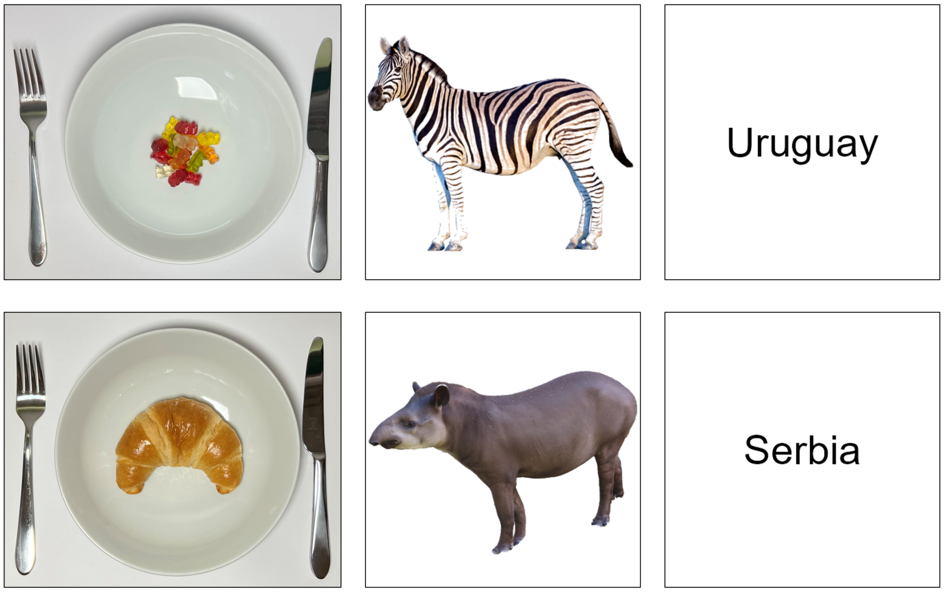
Participants

All participants were recruited via Prolific. Eligibility criteria required participants to (1) be fluent in English, (2) be at least 18 years old, and (3) have an approval rating above 95%. The study took approximately 30 minutes to complete, and participants were compensated £5. Participants could only participate in one domain.

The planned sample size of approximately $N = 50$ participants per domain was determined based on prior model-based estimation studies, in which samples in the range of 30–60 participants are common (e.g., Hoffmann et al., 2016; Juslin et al., 2003; Pachur & Olsson, 2012). Because models were fit and compared at the individual level (see Section

Figure 2

Example stimuli of each domain



Note. Panels display two example stimuli from each domain (left to right): Food (gummy bears and croissants), Mammals (zebra and tapir), and Countries (Uruguay and Serbia)

Modeling), parameter estimation and model discrimination primarily depend on the number of observations per participant rather than solely on the total sample size. We initially recruited $N = 57$ participants for the food domain, $N = 66$ for the country domain, and $N = 59$ for the mammal domain. As preregistered, participants were excluded if they failed at least one of the three attention checks (excluded: $n = 6/14/9$ for food/country/mammal), indicated that their data should not be used for analysis ($n = 0/2/0$), or reported technical problems ($n = 1/2/0$). No participants were excluded based on the other preregistered criteria: (a) mean response times in the testing phase were more than 3 SDs below the sample mean, (b) participants made only two or fewer distinct estimates for all items in the testing phase, (c) they reported using external aids (e.g., looking up answers, taking notes, or asking others), or (d) they provided judgments in under 1000 ms for more than 10% of the items in the estimation phase. Additionally, we excluded all participants who answered fewer than 50% of exemplars correctly (i.e., within ±10% of the true value) in the final training block ($n = 4/0/2$). After exclusions, the final sample comprised $N = 46$ participants in the food domain (age: $M = 40.2$, $SD = 13.9$, range = 19 - 68; 67.4% female), $N = 48$ in the country domain (age: $M = 39.7$, $SD = 12.2$, range = 23 - 66; 43.7% female), and $N = 48$ in the mammal domain (age: $M = 37.5$, $SD = 11.4$, range = 21 - 67; 52.1% female).

At the trial level, we excluded trials with unrealistically high estimates in the testing phase, that is, estimates above 100 for the food and country domain and above 10,000 for the mammal domain ($n_t = 6/2/0$). After these exclusions, the final dataset included $N_t = 3,840$ trials in the food domain, $N_t = 3,840$ in the country domain, and $N_t = 3,674$ in the mammal domain.

Behavioral Results

Analyses were conducted using the statistical software R (Version 4.4, R Core Team, 2024) and Python (Version 3.11.9, Python Software Foundation, 2024). The data, analysis scripts, model files, and intermediate result files are available online (https://osf.io/4m25d/overview?view_only=22adcafffca34c1f8bd10d9835cea98f).

How well did participants learn to estimate real-world quantities?

Since all models assume that people acquire some form of knowledge from previously encountered objects (e.g., cue weights in the rule-based model, exemplars and their criterion values in the exemplar-based model, or categories and their typical criterion values in the Mapping model) we first examined how well participants learned to estimate the criterion values of the exemplars at the end of the training phase. Of course, people might already have some pre-existing knowledge about some of the objects and the corresponding criteria before training. However, there was no significant correlation between estimation accuracy of participants (measured via the root-mean-squared-error, RMSE) and their self-rated knowledge about the domain in the first block of training (food: $r(44) =$ -.03, $p =$ .82;

countries: $r(46)$ = -.03, p = .85; mammals: $r(46)$ = -.25, p = .08) [4]. Still, some of the learning observed in the task may have reflected an activation or refinement of prior knowledge rather than learning entirely from scratch within the experiment.

Overall, participants learned the exemplars in each domain quite well. In the food domain, they correctly[5] estimated an average of 10.7 out of 12 exemplars ($SD = 1.2$, range: 9 - 12) after an average of 6.3 training blocks. In the country domain, participants achieved an average of 11.4 correct estimates ($SD = 0.8$, range: 7 - 12) after 5.1 blocks. In the mammal domain, they reached an average of 10.3 correct estimates ($SD = 1.7$, range: 8 - 12) after 7.0 blocks.

Also, as shown in Figure 3, participants were significantly more accurate when estimating the criterion values of these previously trained (old) stimuli compared to the untrained (new) stimuli in the testing phase, as indicated by the root-mean-squared-error between true and estimated values. This effect was robust across all domains: $t(45) = 16.21$, $p < .001, d = 2.39$ for the food domain, $t(47) = 8.33$, $p < .001, d = 1.20$ for the country domain, and $t(47) = 9.81$, $p < .001, d = 1.42$ for the mammal domain.

How well are people at estimating real-world quantities of untrained (new) stimuli ?

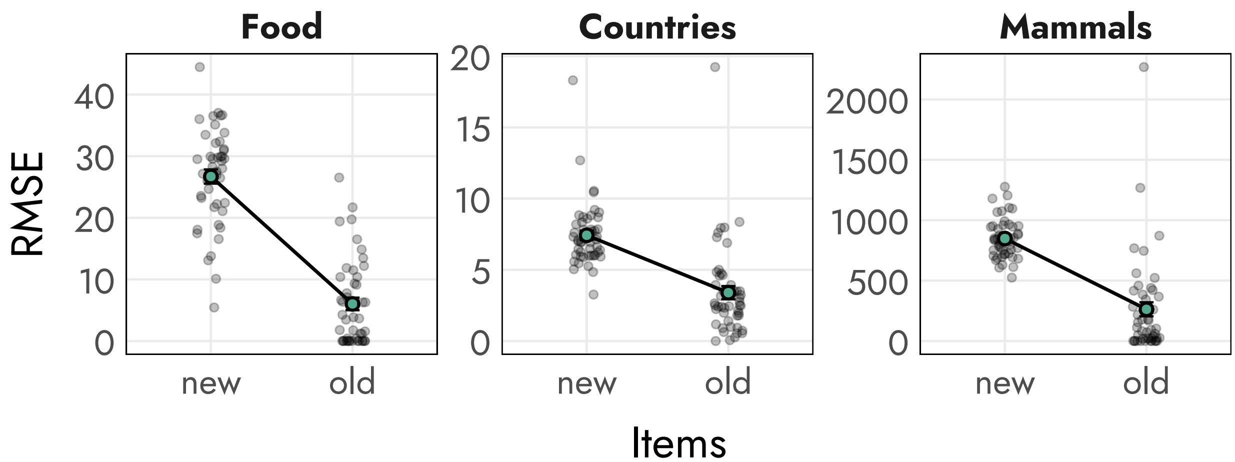
To get a first impression of how accurately participants were able to estimate real-world quantities of untrained (new) stimuli after the training phase, Figure 4A shows the distribution of each participant's estimates alongside the distribution of the actual

[4] In later blocks in the training phase (e.g., Block 5), there were some significant correlations between the RMSE and the self-rated knowledge (food: $r(44)$ = -.30, p = .04; countries: $r(46)$ = -.39, $p < .01$; mammals: $r(46)$ = -.26, p = .07), which might indicate that participants with more pre-existing knowledge are better able to integrate the new information into their existing knowledge. There was, however, no relationship between the RMSE and self-rated knowledge in the testing phase.

[5] A correct estimate was defined as one falling within a $\pm 10\%$ interval of the true criterion value.

Figure 3

Root-mean-squared-error (RMSE) between participants' estimates and the true criterion values for trained (old) and untrained (new) stimuli in the testing phase in each domain.



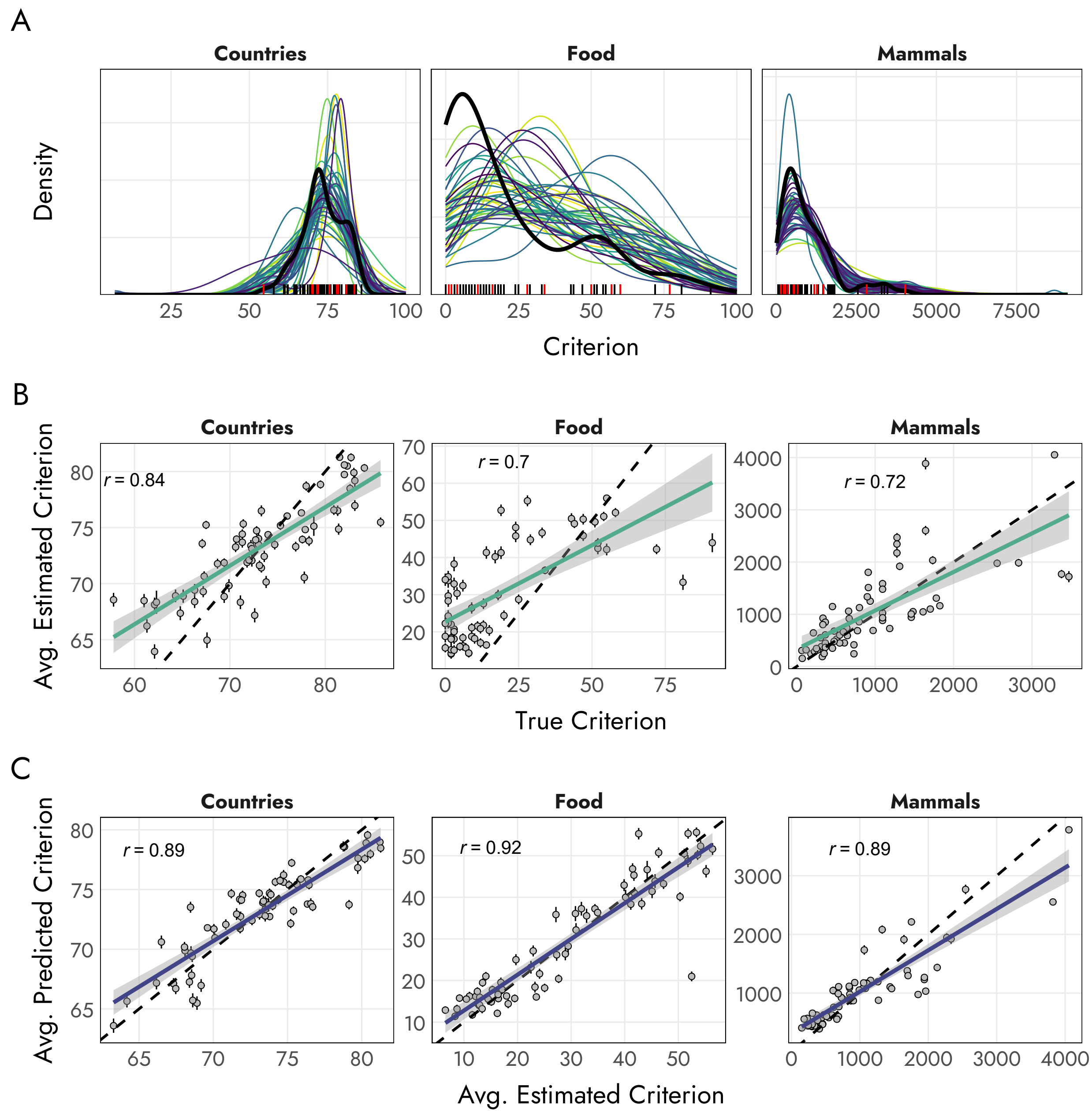
criterion values and Figure 4B the average estimated criterion values for all 68 new stimuli (i.e. stimuli not in the training phase), separated by domain. In addition, Table 1 presents descriptive statistics for the average estimated and true criterion values during the testing phase. Distributions and descriptive statistics indicated that participants' estimates generally aligned well with the actual distributions, particularly in the country and mammal domains. In the food domain, however, participants appeared to overestimate the amount of carbohydrates per 100 grams systematically. This is evident both in the higher mean of the estimated values and in the broader spread compared to the actual values.

Modeling

The behavioral results presented in the previous sections showed that participants successfully learned information about the exemplars and the corresponding domain during the training phase and were able to transfer this knowledge to new stimuli in the testing phase. To evaluate which computational model best captures participants' estimates, we used a simulation-based Bayesian model comparison approach described in the next sections. All analyses were again conducted using the statistical software R (Version 4.4, R Core

Figure 4

Overall estimation performance of participants in each domain.



Note. **A**: Distribution of participants' estimated (in color) and true (in black) criterion values of all 80 stimuli for each domain. Strokes indicate the criterion values of trained (red) and untrained (black) stimuli in each domain. Criteria were grams of carbohydrates per 100g, life expectancy, and days until female maturity for the corresponding domains. **B**: Points represent the average estimated criterion values for each item over all participants. Error bars reflect the corresponding standard error. The dashed line represents perfect estimation (i.e., estimates equal true values). Solid colored lines show the fitted linear regressions relating observed mean estimates to true values. Shaded areas represent 95% confidence intervals of the fitted regressions. **C** Points represent the average median posterior predicted estimated criterion values for each item over all participants (based on the best-fitting model of each participant, y-Axis) by the average actual

Table 1

Descriptive statistics for the true criterion values of stimuli within each domain, participants' estimated values during the testing phase, and the median posterior predicted estimated values. Reported are means (M), standard deviations (SD), and the observed range (Min, Max) for each domain.

	True				Estimated				Median Predicted			
Domain	*M*	*SD*	Min.	Max.	*M*	*SD*	Min.	Max.	*M*	*SD*	Min.	Max.
Food	19.4	21.9	0.0	91.0	30.7	23.6	0.0	100.0	28.9	20.5	0.0	93.1
Countries	73.3	6.6	57.8	85.8	73.3	7.88	7.0	100.0	73.2	6.8	34.5	93.1
Mammals	957.8	772.5	66.0	3,470.0	1,039.8	1,012.1	15.0	9,092.0	1,040.9	791.8	16.3	4,596.1

Note. The criteria were grams of carbohydrates per 100g, life expectancy, and days until female maturity for the corresponding domains. Median predicted values for each participant are based on the model with the highest posterior model probability; more details can be found below.

Team, 2024) and Python (Version 3.11.9, Python Software Foundation, 2024). Analysis scripts and model files are available online (https://osf.io/4m25d/overview?view_only=22adcafffca34c1f8bd10d9835cea98f).

Input Features for the Cognitive Models

All cognitive models described above rely on the same type of input - a set of cues or features used to represent the stimuli - which is then processed in different ways. In our study, the continuous cues, which serve as input to all models, were derived from a multidimensional scaling (MDS) analysis based on a set of pairwise similarity ratings. These ratings were collected in an online study via Prolific, in which participants rated subsets of 158 out of 3,160 possible stimulus pairs across the three domains (food, countries, mammals). After applying exclusion criteria, the final dataset included N = 607 participants for food, N = 520 for countries, and N = 671 for mammals, yielding N_t = 95,393, N_t = 81,103, and N_t = 105,695 pairwise similarity ratings, respectively. A leave-one-out cross-validation procedure determined an optimal dimensionality of the MDS configuration of $m = 14$ cues for the food domain, $m = 10$ for the country domain, and $m = 10$ for the mammal domain. The resulting MDS configurations provided a good reconstruction of the similarity space, with correlations between predicted and observed dissimilarities of $r = .97$, $r = .93$, and $r = .97$, respectively (all ps $< .001$). Further details can be found in the companion paper (Izydorczyk & Bröder, 2025). These MDS coordinates served as continuous input cues for *all* models.

Model Comparison Procedure

To test which of the candidate models best captures participants' estimates of the 68 new stimuli in the testing phase, we used a simulation-based Bayesian model comparison approach. More specifically, we used a recently proposed method of simulation-based inference with deep neural networks as implemented in the `BayesFlow` package (Radev, D'Alessandro, et al., 2023; Radev, Schmitt, et al., 2023) to estimate the posterior model probabilities (PMPs) of all candidate models for each participant separately in each domain[6]. We also included a random guessing model as a baseline in the set of candidate models (see Appendix B for details). The resulting PMPs indicate the probability that each model is the data-generating model, given the observed data and the set of candidate models. The ratio of two PMPs, known as the posterior odds, gives the Bayes Factor assuming equal prior odds. Thus, like Bayes Factors, PMPs encode a notion of Occam's razor as they penalize model complexity (Kass & Raftery, 1995).

The simulation-based method of Bayesian model comparison via `BayesFlow` involves

[6] Initially, we planned (and preregistered) to compare the models on a group level by using hierarchical implementations. However, upon further consideration, we changed to an individual-level comparison in order to be able to also investigate differences in processes between individuals. Nevertheless, the overall model comparison results when hierarchical implementations are used remain consistent with the individual-level result reported below, in that the individual-level model accounting for the largest proportion of participants within a domain was also the hierarchical model with the highest overall posterior model probability for that domain.

the simultaneous training of two neural networks with different roles: a summary network and an inference network (see Figure 5 for a conceptual illustration). The summary network learns maximally informative summary statistics from simulated behavioral data (e.g., judgments, reaction times, choices) and the inference network (i.e., a probabilistic classifier) learns to approximate posterior model probabilities (PMPs) given these learned summaries. The networks are trained on large volumes of simulated data generated from the candidate models, with known generating-model labels. This allows the networks to learn mappings from patterns in the (simulated) data to the underlying models. Once trained, the networks can be used to estimate the PMP of each model based on real data simply by using the real data as input to the network[7].

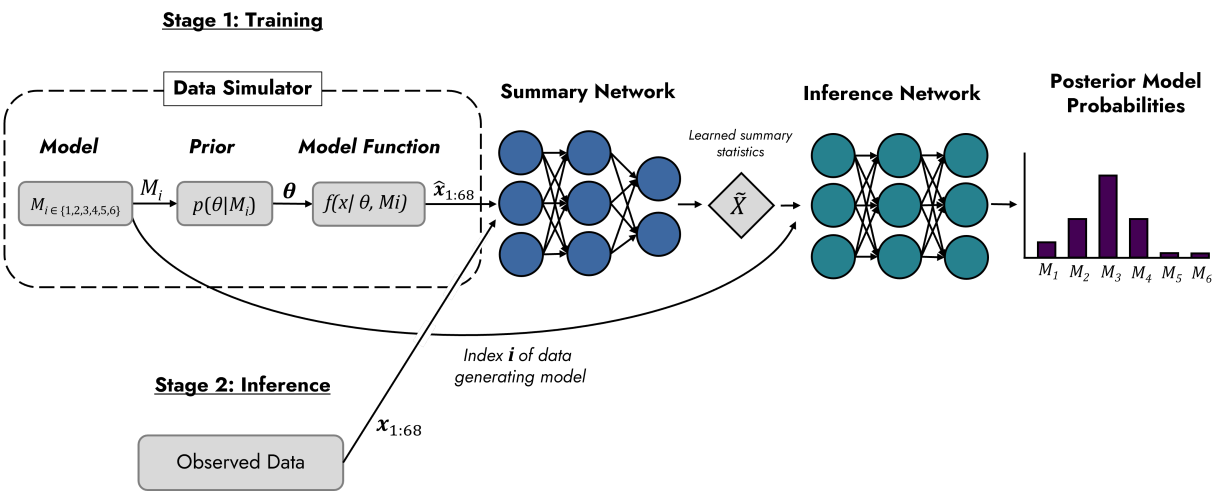
To generate the training data for the networks, we repeatedly simulated data by sampling model parameters from the respective prior distributions defined for the parameters of each candidate model during each iteration of the training process. These priors were chosen to be weakly informative and plausible for each domain. These priors are described in Appendix C. Since the Rule, Exemplar, RulEx-J, and Mapping models as presented above are deterministic given a set of parameters, we assumed a probabilistic response function by modeling the observed estimate y_t on trial t as a draw from a truncated normal distribution with mean $\hat{y}_t$ (the model prediction) and standard deviation σ (capturing response noise). Truncation bounds were domain-specific: 0–100 for the food and country domains, and 0–10,000 for the mammal domain:

$$y_t \sim \text{Normal}_{(0,,100/10{,}000)}(\hat{y}_t, \sigma) \tag{9}$$

[7] A detailed description of `BayesFlow` can be found in Elsemüller et al. (2024), Radev, Schmitt, et al. (2023), and Radev et al. (2022), for application examples see Schumacher et al. (2025), Von Krause et al. (2022), and Wieschen et al. (2024), and for more information about Bayesian modeling in general see Lee and Wagenmakers (2014)

Figure 5

A conceptual illustration of the simulation-based method of Bayesian model comparison via `BayesFlow`



Note. During network training, data of participants' estimates for the 68 stimuli in the testing phase is simulated from the set of 6 candidate models (five process models and one random guessing model) according to the model functions and parameter values drawn from the corresponding prior distributions. The summary network uses the simulated data $\hat{x}$ as inputs and learns maximally informative summary statistics to represent these simulated data. The inference network learns to approximate posterior model probabilities (PMPs) given these summary statistics and information about the true data-generating model. Once trained, the networks can be applied to the observed data x.

Network Training and Validation

Using the approach described above, we trained separate model comparison networks for each of the three domains (food, countries, mammals). Each network was trained over 100 iterations using 32,768 (512 × 64) simulated datasets per training iteration, requiring approximately one hour of training time on a standard CPU. In each simulation step, model parameters were sampled from the respective prior distributions of the candidate models.

To evaluate how well the networks could recover the generating model from simulated data, we validated them on 10,000 simulated datasets per domain. As shown in Figure 6, model classification accuracy was generally high. Some confusion between the RulEx-J model and the Rule (CAM) and Exemplar (GCM) models was observed, which is expected given that both are nested within the RulEx-J model for values of the alpha parameter close to the parameter boundaries. Also, the specific selection of exemplars used may have introduced ambiguity between models for specific parameter combinations.

Figure 6

Accuracy of model classification based on 10,000 simulated datasets

Note. CAM = Cue Abstraction Model, GCM = Generalized Context Model, MAPP = Mapping Model, QEst = Quick-Est Heuristic, RGuess = Random guessing model.

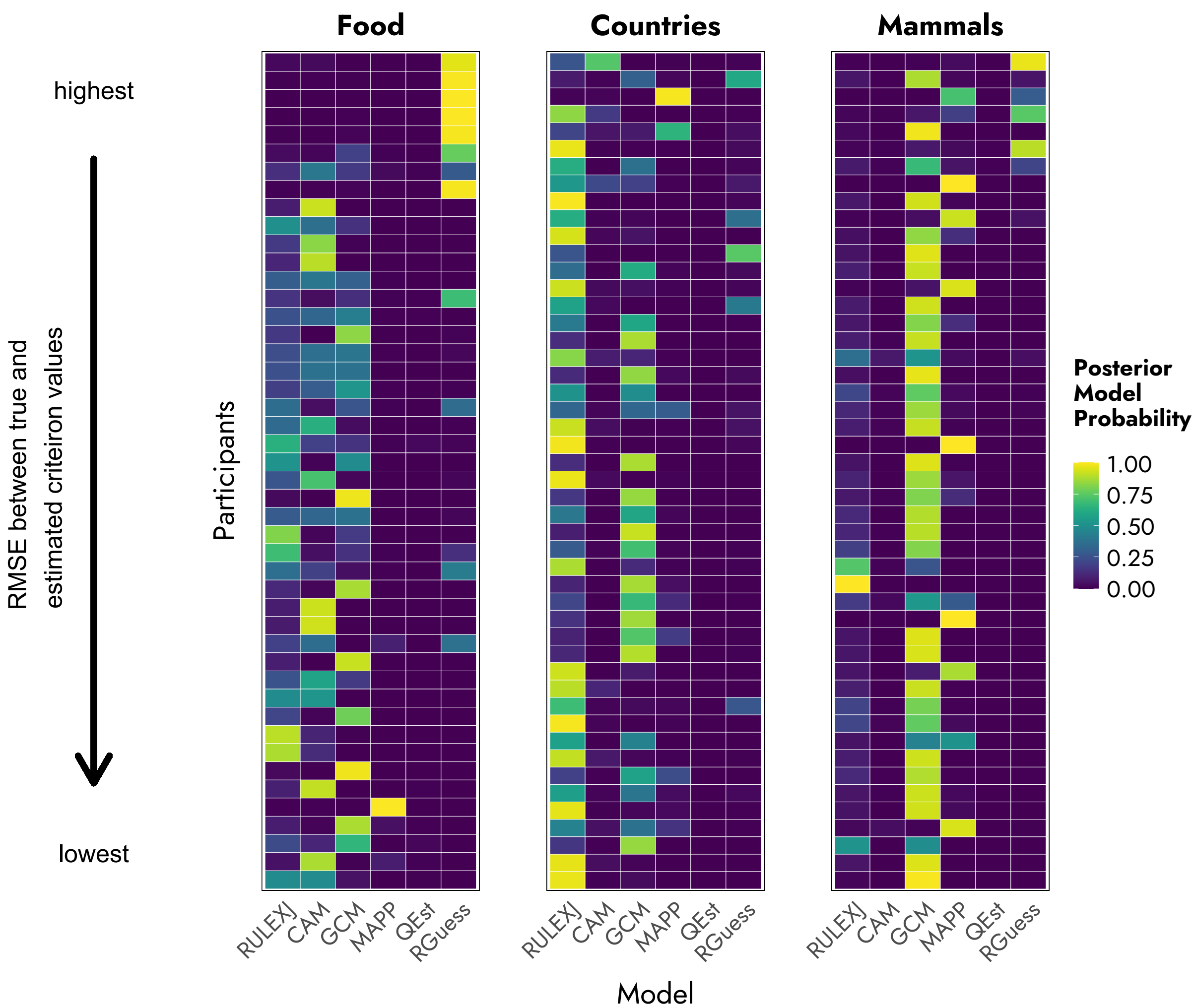
Results

Which process model best captures participants' estimates ?

The resulting posterior model probabilities (PMPs) when participants' estimates were used as input for the corresponding domain-specific model comparison network are shown in Figure 7. Within each domain, participants are ordered by their RMSE between their estimates and the actual criterion values for all 68 test stimuli. Table 2 summarizes the best-fitting model per participant and domain. There are three important results: First, all candidate models (except the Quick-Est model) were able to best account for at least some participants' behavior. Second, in each domain, one or two models clearly dominated in explaining the majority of participants' estimates; for example, in the mammal domain, the Exemplar and the Mapping model had the highest PMP for 54.2% and 33.4% of participants, respectively. Third, the preferred models varied across domains, suggesting that the cognitive processes underlying participants' estimates, as well as the strategies and information used, were at least partly domain-specific. Finally, participants with a high PMP for the random guessing model also tended to have high RMSEs (i.e., low performance), further validating the model comparison approach.

Figure 7

Posterior model probabilities per participant and domain for each of the candidate models.



Note. Participants within each domain are ordered according to the RMSE between their estimates and the actual criterion values of all 68 stimuli in the testing phase. RULEXJ = RulEx-J Model, CAM = Cue Abstraction Model, GCM = Generalized Context Model, MAPP = Mapping Model, QEst = Quick-Est Heuristic, RGuess = Random guessing model.

Table 2

Counts of best fitting models in each domain, determined by the highest posterior model probability for each person.

Domain	RulEx-J	CAM	GCM	MAPP	QEst	RGuess
Food ($n = 46$)	9	13	13	1	0	10
Countries ($n = 48$)	26	1	17	2	0	2
Mammals ($n = 48$)	3	0	33	9	0	3

Note. CAM = Cue Abstraction Model, GCM = Generalized Context Model, MAPP = Mapping Model, QEst = Quick-Est Heuristic, RGuess = Random guessing model.

How well do the process models account for participants' estimates ?

The model comparison results indicated that all candidate models except QuickEst were able to account for some participants' estimation data better than the alternatives. However, these results do not yet clarify how well the models can reproduce participants' actual estimates on average and on an individual level, nor what the best-fitting parameter values are. To address this, we conducted a second set of analyses again using the `BayesFlow` simulation-based inference framework to estimate posterior distributions of individual-level model parameters and generate model-based predictions of each participant's responses based on these parameters. Here, we focused on the four main competitor models: RulEx-J, the rule-based model (CAM), the exemplar-based model (GCM), and the mapping model.

We trained separate networks for each model and each domain, with the networks learning the relationship between simulated estimation data and the corresponding data-generating parameter values. Each network was trained using 32,000 simulated datasets per iteration over 10 to 40 iterations, depending on the number of free parameters of the respective model. Training time ranged from approximately one to two hours per network on a standard CPU. Training success was evaluated by visually inspecting parameter recovery plots, comparing the true and estimated parameters for 300 simulated datasets per model.

We then used the empirical data as input to the trained networks and generated 5,000 posterior samples of parameter values for each participant and, in a subsequent step, generated 5,000 model-based predictions of participants' estimates in all trials based on these posterior samples [8]. Summaries of the best-fitting parameter values can be found in the Tables D1-D7 in the Appendix.

Aggregated Level. To get an impression of how good the models were able to account for participants' estimates on average, we computed the median of the posterior distribution of participants' model predicted estimates in each trial (i.e., the median posterior predictions) for the model that best accounted for the respective participants' data according to the PMP determined in the model comparison step. We excluded participants who were best described by the the Guessing model. Figure 4C shows the average of these median posterior predictions for all new items in the testing phase and the average actual estimates of these items. In addition, Table 1 also includes descriptive statistics for the median posterior predicted estimates. Both show that, on an aggregated level, the posterior predictions from the best-fitting model of each participant account for the actual estimates quite well.

Individual Level. To determine how well the models accounted for participants' estimates on an individual level, we computed R^2 between each participant's actual estimates and the corresponding median posterior predictions generated from their best-fitting model. Overall, the results showed that the model-based predictions explained on average 46.4% of the variance in the food domain, 22.7% in the country domain, and 44.0% in the mammal domain. These values indicate moderate to strong correspondence

[8] As mentioned in the methods section, we excluded eight estimation trials where participants gave unrealistically high estimates (e.g., estimates larger than 100 when estimating the amount of carbohydrates per 100g). However, because the trained networks for parameter estimation require the same shape of data input for all participants (i.e., the same number of trials), we imputed these eight trials by using the average estimated value for this corresponding item. Results did not change when we used different values for the imputation, such as the highest criterion value possible.

between predicted and observed estimates, particularly in the food and mammal domains. However, as evident from the more detailed breakdown in Table 3, the R^2 between participants can vary quite a bit and range from .00 to .89. In addition, Figure 8 shows a negative relationship between each participant's R^2 and their RMSE in the testing phase, that is, between model fit and the accuracy of participants' estimates relative to the true criterion values. This suggests that model performance partially depends on participants' actual estimation accuracy, which may reflect how seriously they engaged with the task.

Table 3

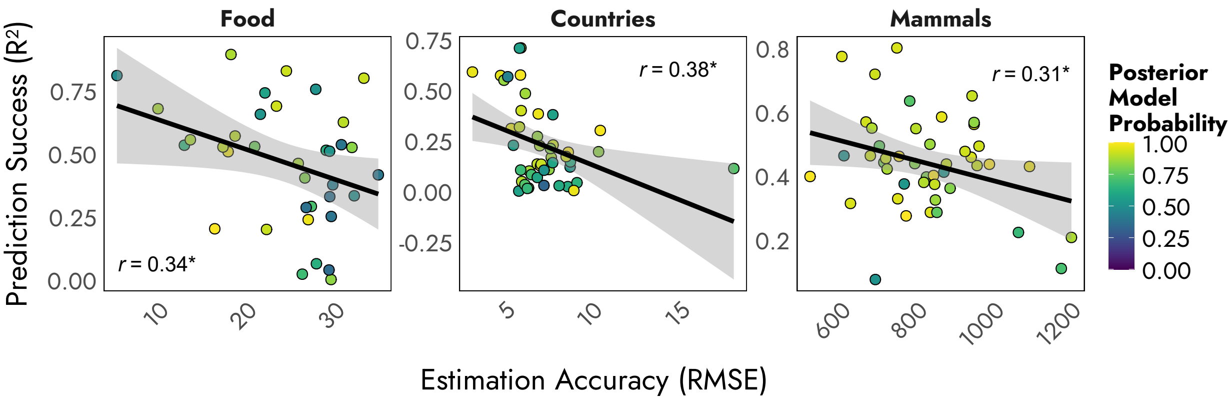
R^2 of actual estimates and median posterior predictions of the best-fitting model of each participant.

Domain	Model	*M*	*SD*	Min.	Max.	*n*
Countries	CAM	0.11		0.11	0.11	1
Countries	GCM	0.14	0.16	0.00	0.55	17
Countries	MAPP	0.17	0.18	0.04	0.30	2
Countries	RULEXJ	0.29	0.21	0.00	0.71	26
Food	CAM	0.60	0.15	0.29	0.83	13
Food	GCM	0.37	0.17	0.00	0.56	13
Food	MAPP	0.20		0.20	0.20	1
Food	RULEXJ	0.43	0.34	0.02	0.89	9
Mammals	GCM	0.46	0.14	0.21	0.80	33
Mammals	MAPP	0.35	0.15	0.08	0.46	9
Mammals	RULEXJ	0.46	0.18	0.27	0.64	3

Note. n = Number of participants who were best described by the corresponding model according to the PMP. CAM = Cue Abstraction Model, GCM = Generalized Context Model, MAPP = Mapping Model, QEst = Quick-Est Heuristic, RGuess = Random guessing model.

Figure 8

Scatterplot of the $\mathbf{R^2}$ *of each participant's actual estimates and the corresponding median posterior predictions and the root-mean-squared-error (**RMSE**) between their estimates and the actual criterion values*



Note. Points are colored based on the posterior model probability (PMP) of the best model of each participant.

General Discussion

To paraphrase a one-liner attributed to the statistician George E. P. Box, the models we use in cognitive science are all wrong, but some are useful. To investigate whether models of quantitative estimation originally developed with artificial and simple stimuli are also useful to account for participants' estimates of real-world stimuli, we collected over 11,000 estimations for a large set of real-world stimuli in three different domains: foods, countries, and mammals. Our findings showed that out of the set of five process models and one guessing model, all candidate models except one (QuickEst) were able to account for a substantial portion of participants' estimation behavior in complex, real-world domains. Across all domains and participants, the exemplar-based models best accounted for the majority of participants' estimates, followed by the RulEx-J model, and then the rule-based and mapping models. Below, we discuss the results regarding general estimation performance, individual differences in model fit, and systematic differences between domains, before turning to a post-hoc interpretation of how structural properties of the environments and criteria may have shaped the relative success of different cognitive process models.

Discrepancies in Estimation Accuracy: Impact of Non-Representative Training Sets

In contrast to the mammals and countries domains, estimation accuracy for food items was significantly lower, characterized by a systematic overestimation of carbohydrate content during the testing phase (see Figure 4). The results may stem from the selection of training exemplars. These items were primarily chosen to maximize discrimination between rule-based and exemplar-based response patterns, and only secondarily with respect to their representativeness of the overall criterion distribution. While training items in the mammals and countries domains closely mirrored the full distribution of values (as evidenced by comparable means, deviations, and ranges), in the food domain the mean carbohydrate content of the training exemplars ($M = 34.75$) was substantially higher than that of the subsequent test items ($M = 19.39$, $t(78) = 2.15$, $p = .037$, $d = 0.66$). Although variability and range were comparable between exemplars and test items, this increased central tendency of the training could have induced an overall elevation in participants' estimates.

This interpretation aligns with the so called "seeding-effect" literature, which shows that providing participants with representative numerical facts about objects in a domain can improve participants' subsequent estimates for these seeded objects and other unseeded objects from the same domain (e.g. Bröder et al., 2023; Brown & Siegler, 1993; Groß et al., 2024). This seeding effect is usually understood as recalibrating and updating people's knowledge about the statistical properties of the criterion (e.g., the distribution, central tendency, and variance) and relative ordering of objects along that criterion. Seeding effects are typically understood as reflecting knowledge updating about a domain's scale and central tendency. Accordingly, the overestimation observed in the food domain can be interpreted as a biased distributional calibration effect induced by the non-representative training set.

Individual Differences in Absolute Model Fit

To assess not only the relative model fits (i.e., which model best accounts for a participant's responses from a set of candidate models) but also the absolute model fit, we conducted an additional analysis in which we generated posterior predictions from the model with the highest posterior model probability for each participant and examined how well these predictions aligned with the participant's actual estimates. On average, the best-fitting model for each participant provided a good account of the average estimates within each domain, see Figure 4C). However, the results revealed substantial individual-level variability in how well the models accounted for participants' actual estimates. One possible explanation for this variability is the degree of systematic variance present in the data. As shown in Figures 7 and 8, a model's ability to account for participants' estimates (as measured by R^2) improved to some degree when participants performed better on the actual estimation task. This is intuitively understandable: models are designed to capture systematic patterns, and when estimates resemble noise or guessing, the models naturally perform worse. Thus, including additional models that implement different guessing strategies or patterns might allow us to capture participants whose responses are difficult to describe with the cognitive process models.

A related limitation of our study is that we only included the base versions of each cognitive model. While this allowed for a clear comparison of core theoretical assumptions, it is important to acknowledge that many of these models have undergone significant development. Various extensions and refinements have been proposed in the literature, for instance, including variable sensitivity parameters in the GCM between exemplars or items (e.g. Nosofsky et al., 2019). Also both the exemplar-based and rule-based models, as well as the RulEx-J hybrid, include mechanisms for feature weighting or cue selection, enabling them to emphasize relevant dimensions while down-weighting irrelevant ones. In contrast, the current implementation of the Mapping model does not incorporate such a mechanism.

Given that cue weighting has proven crucial for the explanatory power of exemplar-based models across various tasks (e.g., Nosofsky, 1986; Rehder & Hoffman, 2005), this omission may have placed the Mapping model at a disadvantage in our comparisons. Incorporating these more advanced and nuanced model variants could potentially enhance their ability to account for individual participant data and might reveal even finer-grained insights into cognitive processes.

Domain Differences in Absolute Model Fit and Structural Determinants of Process Dominance

Besides differences between individuals, the analyses also revealed systematic differences between domains in both the absolute fit of the models (i.e., R^2 between posterior model prediction and participants' estimates) and in the models that most frequently achieved the highest posterior model probability.

Comparison of Explained Variance in Real-World vs. Artificial Stimuli

The average amount of explained variance (as measured by R^2) of participants' estimates differed across domains, with the lowest values for countries, followed by food items, and the highest for mammals, with mean R^2 of .46, .22, and .44 in the food, country, and mammal domains, respectively. The comparatively low explained variance in the country domain may partly reflect the narrower range of sensible criterion values for life expectancy ($SD = 6.6$) compared to carbohydrates in foods ($SD = 21.9$) or time to maturity in mammals ($SD = 772.5$). To further contextualize these values, it may be informative to compare them with results from other quantitative estimation studies using artificial stimuli. For this, we used data from one published study (Izydorczyk & Bröder, 2022) and (re)analyzed data from two unpublished experiments. All three experiments had a similar procedure as the studies reported in this paper, that is, a learning phase where participants learned to judge objects on a continuous criterion based on a small set of training exemplars, followed by a subsequent testing phase with new items. The theoretically best-fitting model for each experiment (e.g., the CAM model for the experiment designed to induce rule-based processing) accounted on average for 83% (SD = 13%, range: 23%–100%), 98% (SD = 3%, range: 85%–100%), and 58% (SD = 16%, range: 13%–88%) of the variance in participants' estimates, with the models being RulEx-J, CAM, and GCM, for the three experiments, respectively. However, these studies involved only small stimulus sets (16–18 items), low-dimensional artificial feature spaces (3–4 cues), very simple linear additive rules connecting cues and criterion values, and only a few items in the training phase (8-10 exemplars), resulting in very high levels of judgment accuracy. In contrast, the present domains involved substantially larger item sets, a more complex cue, and arguably more variable cue structure, prior semantic knowledge, and lower overall estimation accuracy. The lower explained variance observed here, therefore, likely reflects not only model limitations but also the increased structural complexity and noise inherent in realistic environments.

Environmental Structure as a Determinant of Process Dominance

Domain-specific differences also emerged in which models best accounted for participants' behavior. In the mammal domain, most participants were best described by either the exemplar-based or the Mapping model. In the country domain, the RulEx-J and exemplar-based models predominated. In the food domain, the best-fitting models were more evenly distributed across RulEx-J, pure exemplar-based, and pure rule-based models. To interpret this pattern, it is useful to first compare the structural properties of the three domains and criteria. More specifically, the domains and the corresponding criteria differ along at least three dimensions: (a) the strength of taxonomic or categorical organization, (b) the functional form of the cue–criterion relationship (non-linear vs. additive), and (c) the degree to which the environment and cues are perceptually grounded versus semantically abstract.

The mammal domain exhibits a strong taxonomic structure. Stimuli cluster into biologically meaningful families, with high similarity of mammals within categories or families (e.g., visual features, size, habitat), and clear differentiation between them (e.g., apes vs. tigers). Importantly, this categorical organization closely aligns with the criterion: family membership explains 92.7% of the variance in age at female maturity in our sample. Thus, perceptual and taxonomic similarity correspond closely to similarity in criterion values. Beyond this categorical structure, the criterion itself follows a systematic allometric relationship with body size. Despite some exceptions like primates and bats, age at sexual maturity across mammals increases with body size according to robust power-law scaling relationships documented in metabolic and life-history theory (e.g., Blueweiss et al., 1978; Brunet-Rossinni & Austad, 2004; Speakman, 2005; West et al., 1997). Body size, therefore, provides a highly informative cue, but the underlying mapping is fundamentally nonlinear.

In contrast, countries form a more abstract, semantic, and politically shaped domain, weaker categorical organization. Broad groupings such as continents explain substantially less variance in life expectancy ($R^2 = 0.49$). Instead, the life expectancy of a country is strongly predictable from macro-economic and public-health indicators. Across cross-national datasets, linear additive models including variables such as GDP per capita, health expenditure, education, and inequality typically explain a very large proportion of variance in life expectancy (Roffia et al., 2023; Teh et al., 2025). Accurate judgment, therefore, requires access to abstract cue knowledge rather than reliance on perceptual or categorical structure.

The food domain combines two types of properties. On the one hand, carbohydrate content is determined by the natural composition of the food (sugar, starch, and fiber content) and, for processed foods, by recipes and ingredient choices (Menezes et al., 2004). Analytically, carbohydrate content can be calculated by summing individual carbohydrate components (individual sugars, soluble and insoluble dietary fiber contents), and, in principle, it can be approximated through attribute-based reasoning (e.g., grain-based foods tend to contain more carbohydrates than vegetables or meats). On the other hand, foods are organized into salient and meaningful categories or types (e.g., fruits, grains, meats), which also explain a high proportion of variance in carbohydrate content ($R^2 = .69$). Thus, both categorical similarity and attribute-based inference provide informative but incomplete bases for judgment.

The differences in which models best accounted for participants' behavior in these different domains could be in line with the general ecological rationality perspective, which suggests that people adapt their cognitive strategies based on their cognitive capacity, task demands, environmental structure, and feedback about strategy accuracy (e.g., Bröder & Schiffer, 2006; Brusovansky et al., 2018; Gigerenzer et al., 1999; Lee & Gluck, 2021; Marewski & Schooler, 2011; Payne et al., 1988; Rieskamp & Otto, 2006). More specifically, they are consistent with findings from the multiple-cue judgment and categorization literature showing that people rely more on exemplar- or rule-based processes depending on the structure of the stimuli, the task, or environment (e.g., Bröder et al., 2010; Hoffmann et al., 2016; Juslin et al., 2003; Trippas & Pachur, 2019; von Helversen et al., 2013). For example, people tend to rely more on rule-based processes when the environment, the cue format, learning task, or the provided feedback facilitates the abstraction of a (simple) rule (e.g., Bröder et al., 2010; Juslin et al., 2003). Specifically, several studies have shown that when the cue-criterion relationship in an environment is linear, most people make their judgments according to a rule-based process, compared to a non-linear or multiplicative cue-criterion relationship, where people rely more on exemplar-based processes (e.g., Hoffmann et al., 2013, 2014, 2016; Juslin et al., 2008; Karlsson et al., 2007; Olsson et al., 2006). Also, when people have more knowledge about cues (e.g., polarity or relevance), they also tend to rely more on rule-based processes (e.g., Bröder et al., 2010; Platzer & Bröder, 2013; von Helversen & Rieskamp, 2009a; von Helversen et al., 2013). By contrast, exemplar-based processes are favored when the task, the environment, and the stimuli allow participants to form strong representations of specific exemplars. For instance, having more distinguishable exemplars, a lower number of exemplars during training, or more experience have been shown to promote reliance on exemplar-based processing, presumably because these factors increase the strength and discriminability of individual exemplars in memory (e.g., Johansen, 2002; Rouder & Ratcliff, 2004; Thibaut et al., 2018). Based on these findings and purely on post-hoc reasoning, one may sketch the following picture of why the dominant process models differ between domains the way they do and what structural properties of domains and criteria might predict these differences:

Within this framework, the predominance of the exemplar-based and Mapping models in the mammal domain is consistent with the alignment between taxonomic similarity and criterion similarity, combined with a non-linear cue–criterion mapping. Similar species exhibit similar criterion values, making similarity-based generalization effective, while the nonlinear allometric relation limits the usefulness of simple rules.

In the country domain, the dominance of RulEx-J alongside exemplar-based models is consistent with weak categorical structure and incomplete semantic cue knowledge. Because accurate prediction depends on abstract socio-economic indicators that are only partially known, participants may rely on simplified rules derived from incomplete cue representations while simultaneously anchoring judgments to familiar reference countries.

In the food domain, the coexistence of categorical organization and attribute-based structure provides no clear advantage to a single process class or strategy. Category membership supports similarity-based judgments, whereas nutritional reasoning supports rule-based inference. Individual differences in cue knowledge and strategy preference, therefore, yield a heterogeneous distribution of strategies.

The present interpretation is post-hoc, but it yields several testable predictions about when different judgment processes should dominate: Exemplar-based processing should be favored in domains in which perceptual or semantic similarity closely aligns with criterion similarity, particularly when cue–criterion relationships are nonlinear or when other relevant cues are unknown. Rule-based processing should become more prevalent when the criterion can be approximated through a small set of explicitly representable cues and when learners possess reliable knowledge about cue relevance and direction. Finally, heterogeneous strategy use should emerge in environments that simultaneously support categorical similarity and attribute-based inference, or when cue knowledge is incomplete or uneven across stimuli. These predictions suggest that shifts in model dominance should be observable under experimental manipulations that alter category structure, cue transparency, or the availability of cue knowledge.

Leaving the lab or well-designed simplified stimuli always runs the risk to invoke nuisance variables that attenuate the "process-pure" assessment of cognitive strategies or capabilities. The current results make us optimistic that the computational models validated with artificial stimuli also stand the test in the admittedly noisier world of realistic stimuli.

Limitations

A methodological limitation of the present study concerns the use of MDS-based dimensions as model inputs, following recent work in related tasks (Izydorczyk & Bröder, 2024; Nosofsky, Sanders, & McDaniel, 2018; Nosofsky, Sanders, Meagher, & Douglas, 2018; Nosofsky et al., 2019). We believe this approach provides an improvement over hand-coded features, as MDS-derived representations are rooted in human similarity judgments and thus arguably reflect the same mental representations participants rely on in tasks such as categorization, estimation, or memory. However, the method has drawbacks. First, not all features or cues relevant to participants' strategies may be captured in the derived dimensions, particularly since MDS spaces were extracted from similarity ratings provided by a different participant sample (Thibaut et al., 2002). Second, although MDS dimensions are powerful in capturing overall similarity structure, the psychological reality and interpretability of these dimensions can be debated. Future work could extend this approach by combining MDS with external cues (e.g., Hebart et al., 2023; Nosofsky et al., 2019) or alternative data-driven feature extraction methods (e.g., Zeigenfuse & Lee, 2010).

A final limitation of the present study is that, in the naturalistic domains used in this study, participants may have drawn on pre-existing knowledge about the objects and the corresponding quantitative criteria beyond what was provided during training. Such knowledge may include information not captured by the MDS-derived representations or the computational models themselves, and is therefore not reflected in our analyses. Although self-rated domain knowledge was not related to estimation accuracy, the best-fitting model, or the explained variance in estimates, it remains possible that some results were influenced by prior knowledge outside the modeled information. Future research could address this by systematically manipulating or testing the effects of domain expertise, for instance, by comparing participants with differing levels of expertise.

Implications and Conclusion

This study shows that established computational models of quantitative estimation, long tested with simplified laboratory stimuli, can also capture people's judgments in naturalistic domains. By applying established models to real-world stimuli from the food, country, and mammal domains, we demonstrate that the cognitive mechanisms underlying estimation generalize beyond controlled experimental settings, showing that, perhaps, even the spherical cow in the vacuum can teach us something about the real world.

At the same time, the results reveal clear domain dependencies, where different models best accounted for most participants' estimates in each domain. This pattern might indicate that people flexibly adapt their estimation strategies to the informational structure of the environment. Future research should build on this domain dependency of processes to examine which aspects of a domain determine what information becomes relevant for different cognitive mechanisms.

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Appendix A

Item Lists

Table A1

Eighty food items with gramm carbohydrates per 100g.

Item		Item		Item		Item	
Spaghetti	**28**	Cauliflower	2	Gouda cheese	0	Chocolate	55
Rice	28	**Mushrooms**	**1**	Butter	1	Vanilla pudding	14
Rigatoni	34	Peas	9	Feta	1	Chips	51
Bread roll	44	Lentils	25	Mozzarella	1	Pretzel sticks	72
Oat flakes	**60**	Beans	3	Cream cheese	3	**Apple turnover**	**34**
Couscous	29	Banana	20	Salami	1	**Chocolate cake**	**50**
Bread	43	Apple	14	Dried ham	1	Cheesecake	33
Croissant	47	Watermelon	8	Vienna sausages	1	Cookies	58
White toast	47	Kidney beans	13	Tuna (canned)	0	Peanuts	11
Bulgur	24	Orange	8	Shrimps	3	**Walnuts**	**4**
Wrap	52	Mandarin	10	Salmon	0	Whipped cream	3
Potatoes	19	Mango	13	Pork	0	**Chickpeas**	**16**
Cucumber	2	Pineapple	12	**Chicken**	**2**	Radishes	2
Bell pepper	6	Strawberry	6	**Nutella**	**57**	Fish sticks	18
Carrots	7	Raspberry	5	Cherry jam	55	Vegetarian salami	9
Corn	**11**	Pear	12	Raspberry jam	52	French fries	24
Fried egg	2	Apricot	9	Honey	81	Kinder chocolate bar	54
Broccoli	2	Grapes	15	Peanut butter	17	Eggplant	3
Tomatoes	3	Yogurt	6	**Gummy bears**	**77**	**Rice cake**	**77**
Iceberg lettuce	2	Curd cheese	4	Ice cream	19	Dextrose	91

Note. Items learned during the training phase are shown in bold.

Table A2

Eighty country items with life expectancy.

Item		Item		Item		Item	
China	78	Angola	64.8	United Arab Emirates	83.1	Suriname	73.8
India	72.2	Peru	77.9	Libya	71.1	**Brunei**	**75.5**
United States of America	79.5	Yemen	69.4	El Salvador	72.3	Iceland	83
Nigeria	**54.6**	Ivory Coast	62.1	Serbia	76.9	Maldives	81.3
Brazil	76	Nepal	70.6	Paraguay	74	Sao Tome & Principe	69.9
Russia	73.3	Venezuela	72.7	Ireland	82.6	Samoa	71.8
Mexico	75.3	Australia	84.1	New Zealand	82	Saint Lucia	72.8
Ethiopia	67.6	Taiwan	80.9	**Costa Rica**	**81**	Saint Vincent & the Grenadines	71.4
Egypt	**71.8**	Syria	72.6	Croatia	78.8	Seychelles	73
Vietnam	74.7	Burkina Faso	61.3	**Mongolia**	**72**	Tonga	73.1
Democratic Republic of the Congo	62.1	Chile	81.4	**Uruguay**	**78.3**	Dominica	71.3
Germany	81.5	Netherlands	82.3	**Jamaica**	**71.6**	Saint Kitts & Nevis	72.3
Iran	77.8	Ecuador	77.6	Moldova	71.3	Marshall Islands	67.1
France	**83.5**	Benin	61	Lesotho	57.8	San Marino	85.8
South Africa	66.3	Bolivia	68.7	Guinea-Bissau	64.3	Palau	69.4
Tanzania	67.2	Papua New Guinea	66.3	Trinidad & Tobago	73.6	**Cook Islands**	**75.5**
Colombia	**77.9**	Haiti	65.1	Fiji	67.5	Nauru	62.3
Argentina	77.5	Cuba	78.3	Bhutan	73.3	Tuvalu	67.3
Canada	82.7	Greece	82	**Guyana**	**70.3**	**Niue**	**70.1**
Ukraine	72	Honduras	73	Solomon Islands	70.7	Vatican City	83.1

Note. Items learned during the training phase are shown in bold.

Table A3

Eighty mammal items with days until female maturity.

Item		Item		Item		Item	
Aoudad	335	Raccoon dog	304	**Western gray kangaroo**	**670**	Japanese macaque	1483
Brazilian tapir	1095	Bat-eared fox	365	Quokka	252	Gorilla	2829
American bison	912	Cheetah	456	**Sugar glider**	**236**	Chimpanzee	3376
Giant anteater	1186	Lion	1095	Brush-tailed possum	315	Orangutan	2555
Ibex	797	Leopard	937	Koala	646	Brown lemur	509
Thomson's gazelle	365	**Tiger**	**1268**	Mountain hare	266	Black lemur	540
Llama	800	Dwarf mongoose	456	European rabbit	730	Ring-tailed lemur	595
Kirk's dik-dik	213	Meerkat	365	Duck-billed platypus	548	**Slender loris**	**380**
Red panda	**550**	Spotted hyena	1095	Short-beaked echidna	548	white-faced saki	775
African buffalo	**1475**	Sea otter	974	Horse	914	**African bush elephant**	**4018**
Bactrian camel	1278	Least weasel	120	Quagga	900	Naked mole-rat	228
Dromedary	1095	Steller sea lion	1780	White rhinoceros	1643	Guinea pig	66
Mule deer	478	Harbor seal	1095	Central American spider monkey	1825	Degu	182
Reindeer	662	Common raccoon	365	White-tufted-ear marmoset	477	Lesser Egyptian jerboa	327
Giraffe	1278	Brown bear	1313	Midas tamarin	608	Long-tailed field mouse	71
Hippopotamus	1279	Polar bear	1734	Tufted capu	1703	**Nutria**	**152**
Wart hog	**578**	Humpback whale	1643	De Brazza's monkey	1611	**Eurasian red squirrel**	**296**
Wild boar	334	**Bottlenosed dolphin**	**2831**	Guereza	1461	Siberian chipmunk	350
Gray wolf	669	Australian tiger cat	340	**Stump-tailed macaque**	**1186**	Dugong	3470
African wild dog	639	Tasmanian devil	730	Proboscis monkey	1460	Asiatic elephant	3287

Note. Items learned during the training phase are shown in bold.

Appendix B

Random Guessing Model

The baseline random guessing model assumes that participants learned the smallest ($\min_{crit_e}$) and largest criterion value ($\max_{crit_e}$) of the exemplars encountered during training with some precision σ_{RGuess}:

$$\begin{aligned} \hat{\min}_{crit_e} &\sim \text{Normal}(\min_{crit_e}, \sigma_{RGuess}) \\ \hat{\max}_{crit_e} &\sim \text{Normal}(\max_{crit_e}, \sigma_{RGuess}) \end{aligned} \tag{B1}$$

with the constraint that $\hat{\min}_C < \hat{\max}_C$. Participants' estimates for a probe p are then assumed to be random samples from a uniform distribution with the learned smallest ($\hat{\min}_C$) and largest ($\hat{\max}_C$) criterion value as lower and upper bounds:

$$y_p \sim \text{Uniform}(\hat{\min}_C, \hat{\max}_C) \tag{B2}$$

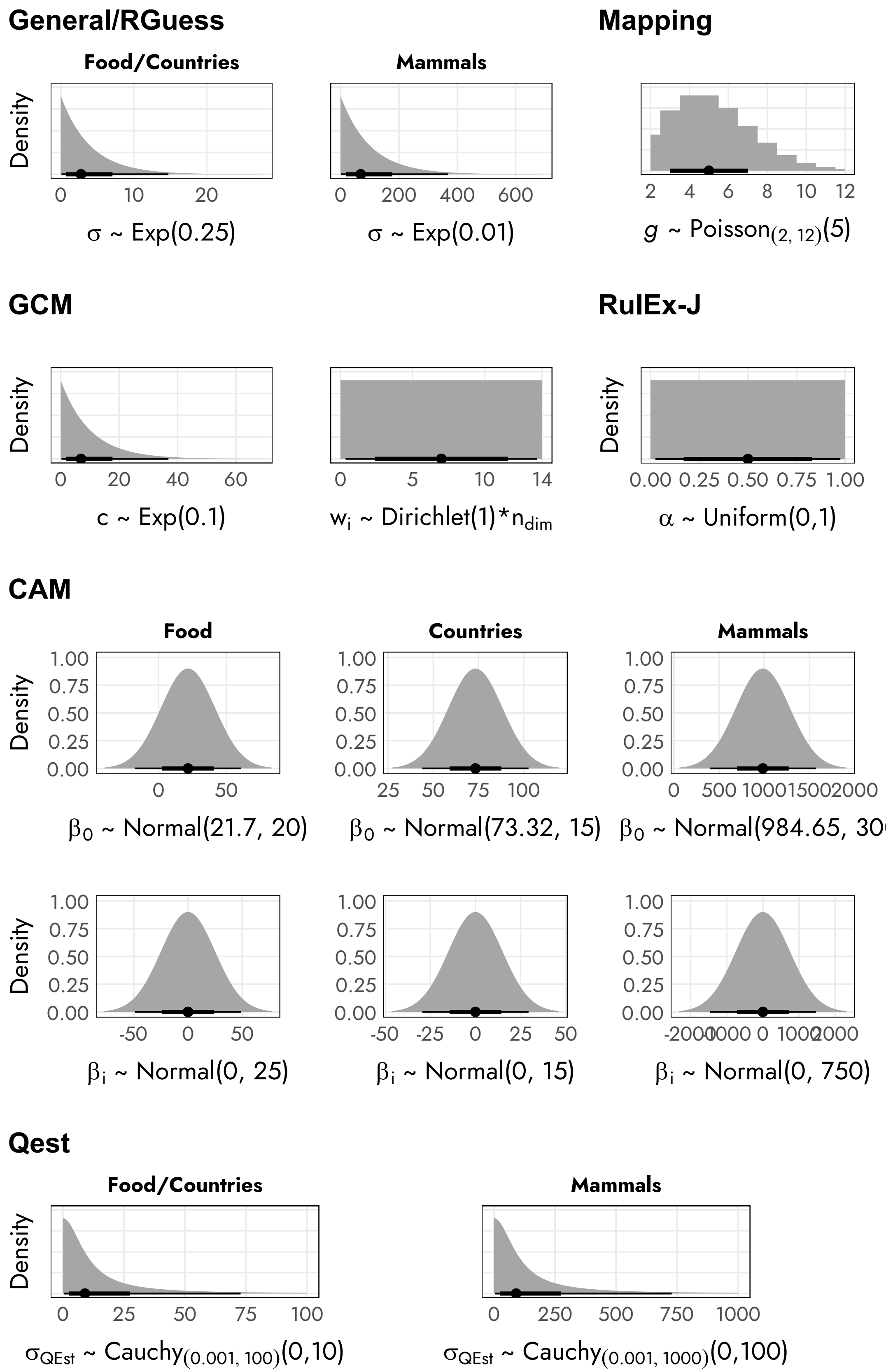
Appendix C

Prior Distributions

We used weakly informative, plausible prior distributions for all model parameters (see Figure C1). Prior plausibility was assessed via prior predictive checks, where we repeatedly simulated and plotted predicted responses for 50 participants based on random draws from the respective priors. For the intercept (β_0) and cue weights (β_i), priors were guided by the distribution of linear regression coefficients obtained when predicting the actual criterion values from the cues. For the RulEx-J model, the priors for the rule- and exemplar-based submodules were specified identically to those used in the corresponding standalone models.

Figure C1

Prior distribution for all model parameters.



Appendix D

Estimated Parameters

The following tables summarize the posterior parameter estimate results for models that provided the best fit for at least five participants. For each model, we report the average posterior parameter values, including the mean, standard deviation, and a 95% Highest Density Interval (HDI). The HDI is a measure of the most credible range of parameter values. The "best-fitting" model for each participant was determined by the highest Posterior Model Probability (PMP), as detailed in the main text.

Food
Table D1

Average posterior parameter values of the ***CAM*** *model of* $N = 13$ *participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
β_0	29.04	1.81	25.40	32.03
β_1	-47.77	5.29	-58.32	-37.62
β_2	-12.81	8.49	-28.66	3.88
β_3	-43.39	6.51	-56.51	-30.83
β_4	-11.59	7.21	-26.03	2.54
β_5	2.83	10.93	-18.53	23.92
β_6	4.38	12.15	-19.48	28.08
β_7	5.44	13.55	-20.70	32.32
β_8	0.78	22.63	-43.34	43.14
β_9	-1.01	19.94	-39.25	37.45
β_{10}	-5.11	18.92	-42.30	31.21
β_{11}	7.96	19.15	-29.28	45.02
β_{12}	-2.64	21.93	-45.66	40.24
β_{13}	3.83	21.65	-38.07	46.49
β_{14}	12.55	20.81	-27.19	53.55
σ	4.99	2.99	0.00	10.31

Table D2

*Average posterior parameter values of the **GCM** model of N = 13 participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
c	35.92	10.72	17.78	57.45
w_1	1.80	0.89	0.39	3.55
w_2	0.34	0.28	0.00	0.85
w_3	0.59	0.49	0.01	1.53
w_4	1.83	0.98	0.30	3.72
w_5	0.94	0.61	0.04	2.07
w_6	1.08	0.99	0.00	3.03
w_7	0.77	0.71	0.04	2.18
w_8	1.03	0.78	0.00	2.45
w_9	0.80	0.71	0.00	2.20
w_{10}	0.85	0.75	0.00	2.34
w_{11}	1.23	0.89	0.01	2.92
w_{12}	0.57	0.48	0.00	1.48
w_{13}	1.29	0.80	0.04	2.76
w_{14}	0.94	0.77	0.00	2.43
σ	8.50	2.23	4.35	12.96

Table D3

*Average posterior parameter values of the **RULEX-J** model of N = 9 participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
α	0.43	0.17	0.11	0.76
$\beta_{0\ \text{CAM}}$	15.93	10.44	-5.18	32.03
$\beta_{1\ \text{CAM}}$	-23.23	20.17	-62.73	18.29
$\beta_{2\ \text{CAM}}$	13.82	17.47	-19.88	49.46
$\beta_{3\ \text{CAM}}$	-19.32	18.36	-55.43	17.98
$\beta_{4\ \text{CAM}}$	-20.05	20.41	-59.60	21.67
$\beta_{5\ \text{CAM}}$	-12.31	18.33	-48.97	23.77
$\beta_{6\ \text{CAM}}$	-16.99	18.85	-54.55	20.39
$\beta_{7\ \text{CAM}}$	2.86	20.05	-36.92	42.29
$\beta_{8\ \text{CAM}}$	1.67	20.68	-39.55	42.82
$\beta_{9\ \text{CAM}}$	4.15	19.13	-33.49	42.22
$\beta_{10\ \text{CAM}}$	5.57	21.72	-36.60	49.41
$\beta_{11\ \text{CAM}}$	-0.50	20.25	-40.44	39.61
$\beta_{12\ \text{CAM}}$	1.09	21.93	-41.75	44.81
$\beta_{13\ \text{CAM}}$	-10.95	20.86	-52.00	30.16
$\beta_{14\ \text{CAM}}$	11.50	20.96	-29.92	52.59
c	18.34	11.60	3.87	40.20
$w_{1\ \text{GCM}}$	1.21	1.00	0.02	3.20
$w_{2\ \text{GCM}}$	0.53	0.59	0.00	1.72
$w_{3\ \text{GCM}}$	0.92	0.85	0.01	2.64
$w_{4\ \text{GCM}}$	1.04	0.96	0.00	2.97
$w_{5\ \text{GCM}}$	1.29	0.96	0.06	3.20
$w_{6\ \text{GCM}}$	0.99	0.98	0.00	2.99
$w_{7\ \text{GCM}}$	1.43	1.04	0.14	3.47
$w_{8\ \text{GCM}}$	1.26	0.98	0.02	3.17
$w_{9\ \text{GCM}}$	0.72	0.71	0.00	2.14
$w_{10\ \text{GCM}}$	0.82	0.79	0.00	2.43
$w_{11\ \text{GCM}}$	1.14	0.98	0.00	3.12
$w_{12\ \text{GCM}}$	0.87	0.75	0.00	2.35
$w_{13\ \text{GCM}}$	0.99	0.86	0.00	2.72
$w_{14\ \text{GCM}}$	0.86	0.89	0.00	2.68
σ	11.81	2.43	7.24	16.54

Countries
Table D4

*Average posterior parameter values of the **GCM** model for $N = 12$ participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
c	6.29	2.94	1.75	11.76
w_1	0.99	0.70	0.04	2.36
w_2	0.90	0.77	0.01	2.44
w_3	0.74	0.70	0.00	2.15
w_4	0.89	0.80	0.00	2.49
w_5	0.92	0.82	0.00	2.58
w_6	1.22	0.94	0.01	3.03
w_7	1.28	1.00	0.04	3.24
w_8	1.29	0.95	0.00	3.12
w_9	0.73	0.67	0.00	2.08
w_{10}	1.05	0.86	0.01	2.75
σ	4.75	0.68	3.47	6.10

Table D5

Average posterior parameter values of the ***RULEX-J*** *model for $N = 24$ participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
α	0.30	0.12	0.08	0.55
$\beta_{0\ \mathrm{CAM}}$	72.22	5.97	61.28	82.31
$\beta_{1\ \mathrm{CAM}}$	-13.97	11.63	-37.28	8.96
$\beta_{2\ \mathrm{CAM}}$	1.13	10.02	-18.20	21.69
$\beta_{3\ \mathrm{CAM}}$	9.20	8.90	-7.88	27.46
$\beta_{4\ \mathrm{CAM}}$	-0.88	9.62	-20.26	18.08
$\beta_{5\ \mathrm{CAM}}$	8.26	9.47	-10.20	27.39
$\beta_{6\ \mathrm{CAM}}$	5.01	9.94	-14.61	24.95
$\beta_{7\ \mathrm{CAM}}$	-4.25	10.42	-25.23	16.30
$\beta_{8\ \mathrm{CAM}}$	2.53	10.57	-18.73	23.51
$\beta_{9\ \mathrm{CAM}}$	-8.14	10.67	-29.68	12.65
$\beta_{10\ \mathrm{CAM}}$	2.28	11.23	-20.12	24.91
c	6.65	5.29	0.50	16.38
$w_{1\ \mathrm{GCM}}$	1.30	0.97	0.04	3.24
$w_{2\ \mathrm{GCM}}$	1.04	0.91	0.00	2.89
$w_{3\ \mathrm{GCM}}$	1.08	0.93	0.00	2.94
$w_{4\ \mathrm{GCM}}$	0.81	0.78	0.00	2.39
$w_{5\ \mathrm{GCM}}$	0.93	0.84	0.00	2.63
$w_{6\ \mathrm{GCM}}$	0.96	0.85	0.00	2.68
$w_{7\ \mathrm{GCM}}$	0.96	0.85	0.00	2.68
$w_{8\ \mathrm{GCM}}$	1.00	0.89	0.00	2.80
$w_{9\ \mathrm{GCM}}$	1.01	0.83	0.01	2.67
$w_{10\ \mathrm{GCM}}$	0.89	0.81	0.00	2.52
σ	5.00	0.51	4.03	6.02

Mammals
Table D6

Average posterior parameter values of the ***GCM*** *model of $N = 33$ participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
c	37.94	207.58	14.33	60.66
w_1	0.71	0.51	0.03	1.67
w_2	1.25	0.85	0.13	2.89
w_3	0.52	0.41	0.02	1.30
w_4	0.77	0.60	0.00	1.91
w_5	0.94	0.79	0.01	2.46
w_6	1.07	0.67	0.15	2.37
w_7	1.14	0.77	0.03	2.57
w_8	1.21	0.81	0.00	2.68
w_9	1.33	0.68	0.23	2.63
w_{10}	1.07	0.79	0.07	2.58
σ	314.58	111.53	183.09	458.00

Table D7

Average posterior parameter values of the ***MAPPING*** *model of $N = 9$ participants.*

Parameter	M	SD	HDI-2.5%	HDI-97.5%
g	5.02	1.41	2.01	7.57
σ	502.65	121.93	289.14	755.20